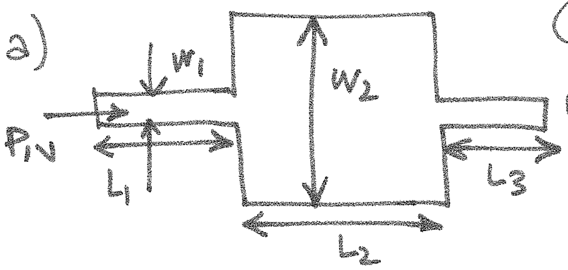
EXERCISE 1

a) (NOT TO SCALE)



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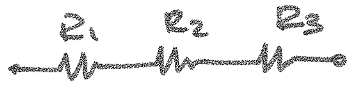


$\frac{W}{h} > 1.4 \longrightarrow \qquad R = \frac{12\,\eta\, L}{\left(1-0.63\,\frac{h}{W}\right) h^3 \cdot W}$

$R_3 = R_1/2$

$R_1 = \frac{12 \cdot \eta\, L_1}{\left(1-0.63\,\frac{H}{W_1}\right) h^3 \cdot W_1} = \frac{12 \cdot 10^{-3} \cdot 2\,cm}{\left(1-0.63\,\frac{40}{200}\right)(40\cdot 10^{-6})^3 \cdot 200 \cdot 10^{-6}} = 21.4 \cdot 10^{12} \left[\frac{Pa\cdot s}{m^3}\right]$

$R_3 = 10.7 \cdot 10^{12}\,[Pa\cdot s/m^3]$

$R_2 = \frac{12 \cdot \eta\, L_2}{\left(1-0.63\,\frac{h}{W_2}\right) \cdot h^3 \cdot W_2} = \frac{12 \cdot 10^{-3} \cdot 1\,cm}{\left(1-0.63\,\frac{40}{1000}\right)(40^{-6})^3 \cdot 1000e^{-6}} = 2 \cdot 10^{12} \left[\frac{Pa\cdot s}{m^3}\right]$

$R_{TOT} = R_1 + R_2 + R_3 = 34.1 \cdot 10^{12}\,[Pa\cdot s/m^3]$

$Q = \frac{\Delta P}{R} = \frac{110\,kPa - 101\,kPa}{34.1 \cdot 10^{12} \left[\frac{Pa\cdot s}{m^3}\right]} =$

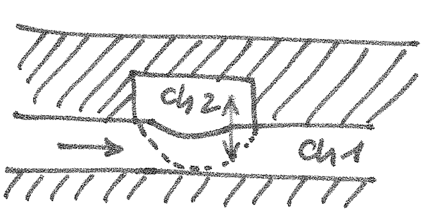
$= \frac{9\,kPa}{34.1 \cdot 10^{12} \ldots} = \boxed{264 \cdot 10^{-12}\, m^3/s}$

$(1\,m^3 = 1000\,\ell)$ $\qquad$ FLOW-RATE $\qquad$ $\boxed{(15.8\,\mu\ell/min)}$

$V_1 = \frac{Q}{W_1 \cdot H} = \frac{264 \cdot 10^{-12}\,\frac{m^3}{s}}{40\,\mu m \cdot 200\,\mu m} = 33\,mm/s$

$V_2 = \frac{Q}{W_2 \cdot H} = \frac{V_1}{5} = 6.6\,mm/s$

b)



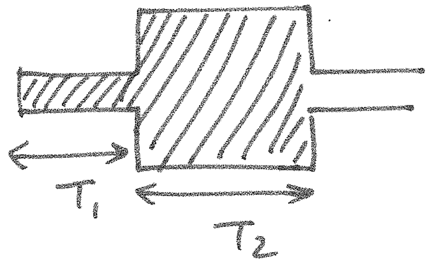
Pressure in channel 2 controls the flexible membrane that opens/closes channel 1.

c) $\text{Volume}\big|_{PCR\_chamber} = W_2 \cdot H \cdot L_2 =$

$= 40\,\mu m \cdot 2000\,\mu m \cdot 1\,cm = 400 \cdot 10^{-12}\,m^3$

$= (0.4\,\mu l)$

Filling time: $T_{TOT} \sim T_1 + T_2$

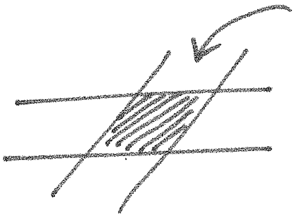


$T_1 = \frac{L_1}{V_1} = \frac{2\,cm}{33\,mm/s} = 0.6$ sec

$T_2 = \frac{L_2}{V_2} = \frac{1\,cm}{6.6\,mm/s} = 1.51$ sec

$T_{TOT} = 2$ sec. ( )

d) flexible



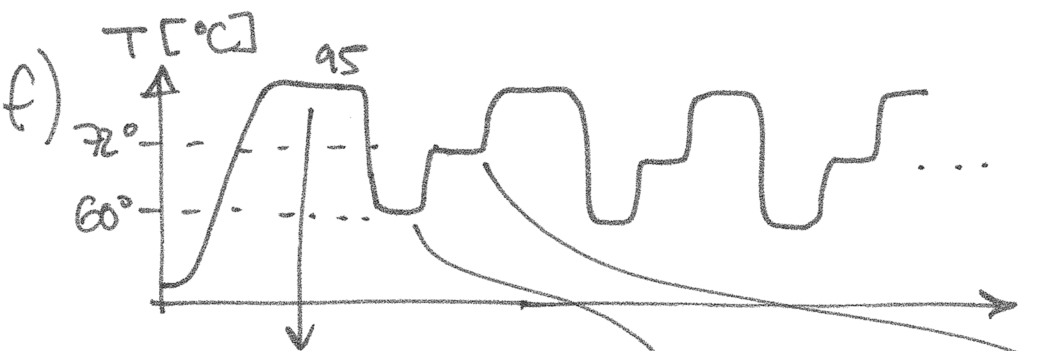
$C_H = \frac{\text{Area}^2}{K} = \frac{(200\,\mu m \cdot 200\,\mu m)^2}{20\,N/m}$

$\boxed{8 \cdot 10^{-17}\ m^3/Pa}$

$\tau = \underbrace{(R_1 + R_2)}_{7.3 \cdot 10^{12}} /\!/ R_3 \cdot \underbrace{C_H}_{8 \cdot 10^{-17}} = 0.6\,ms$ (FAST!)

e) $T_{max} = T_{DNA\ denaturation}$ 95°C

f)



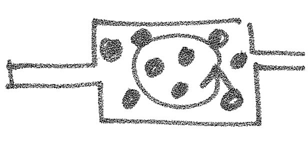
DENATURATION: (strands separation)

ANNEALING:

PRIMER

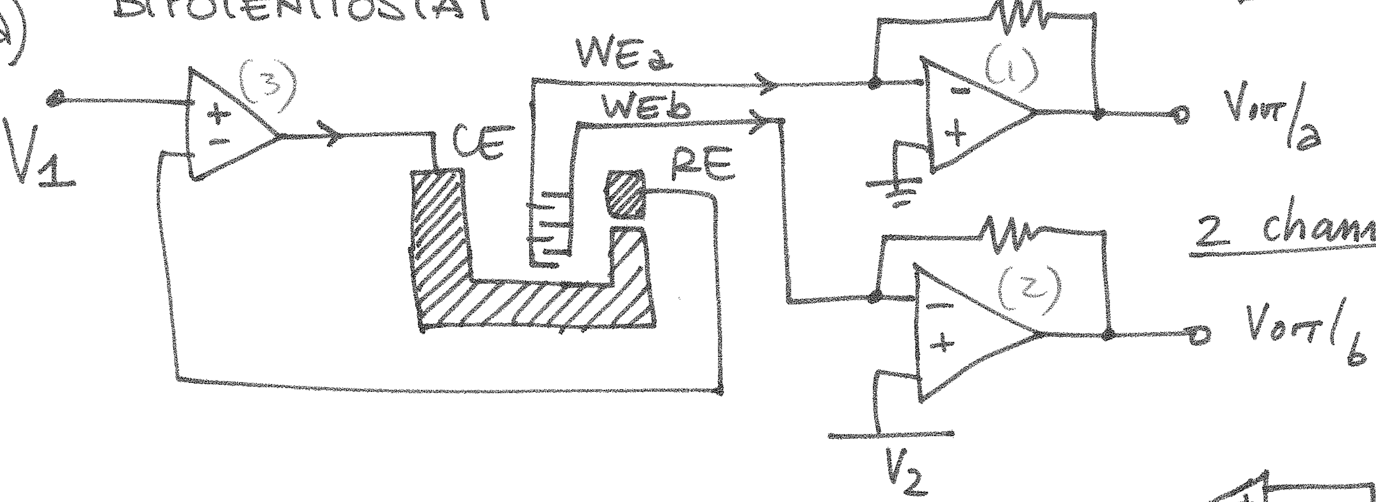
EXTENSION: enzyme operation to connect nucleotides

g) MAGNETIC BEADS INSIDE THE CHAMBER (ROTATING MAGNET)



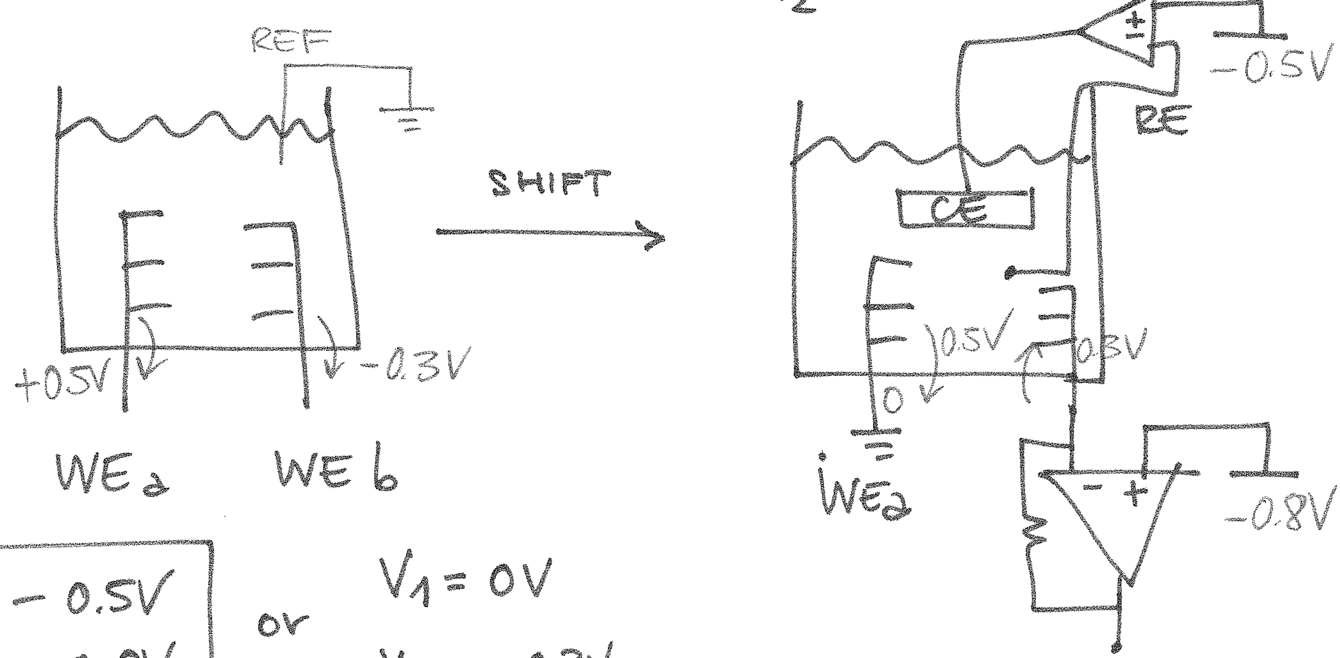
EXERCISE 2

a) BIPOTENTIOSTAT



2 channels

b)



$$\boxed{\begin{aligned} V_1 &= -0.5V \\ V_2 &= -0.8V \end{aligned}}$$

or

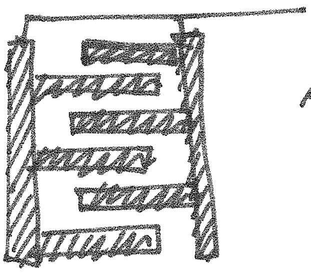
$V_1 = 0V$

$V_2 = -0.3V$

($V_x = +0.5V$ VIRTUAL GROUND 1st opamp)

c) $C_{DL} = \text{AREA} \cdot C_0$ (IN PARALLEL)

AREA :

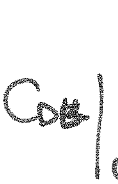


$$A = 6 \cdot L \cdot W + 2 \cdot H \cdot W =$$

$$= W(6L + 2H) = 5(6 \cdot 50 + 2 \cdot 60)$$

$$= 2100\,\mu m^2$$

$$C_{DL}\Big|_{WE} = 2100\,\cancel{\mu m^2} \cdot 0.1 \frac{pF}{\cancel{\mu m^2}} = \underline{210pF}$$



$$C_{DL}\Big|_{CE} = \text{AREA}_{CE} \cdot C_0 = T \cdot P + Z \cdot P + (Q - 2T) \cdot T$$

$$= T(P + Z + Q - 2T) =$$

$$= 30(100 + \cancel{60} + 100 - \cancel{60}) = 6000\,\mu m^2$$

3 times $C_{DL}|_{WE}$

d) Thermal noise:

$$\frac{4KT}{R} \longrightarrow \sigma_{noise} = \sqrt{\frac{4KT}{R} \cdot \frac{\pi}{2} \cdot BW} = 100\ pA_{RMS}$$

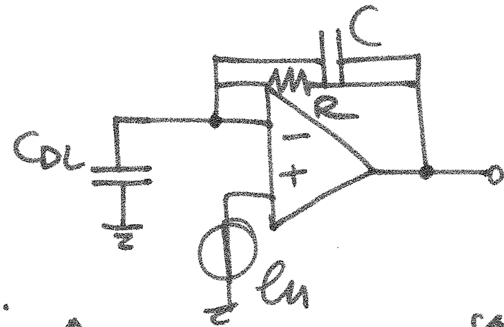
$$\sqrt{\frac{4KT}{9} \cdot \frac{9}{R}} = 0.\cancel{4}\ pA/\sqrt{Hz}$$

$$0.2\ pA/\sqrt{Hz} \cdot \sqrt{\frac{\pi}{2} \cdot BW} = 100\ pA \Longrightarrow BW = \left(\frac{100\ pA}{\cancel{\sqrt{Hz}}\ 0.\cancel{4}pA/\sqrt{Hz}} \cdot \frac{2}{\pi}\right)^2$$

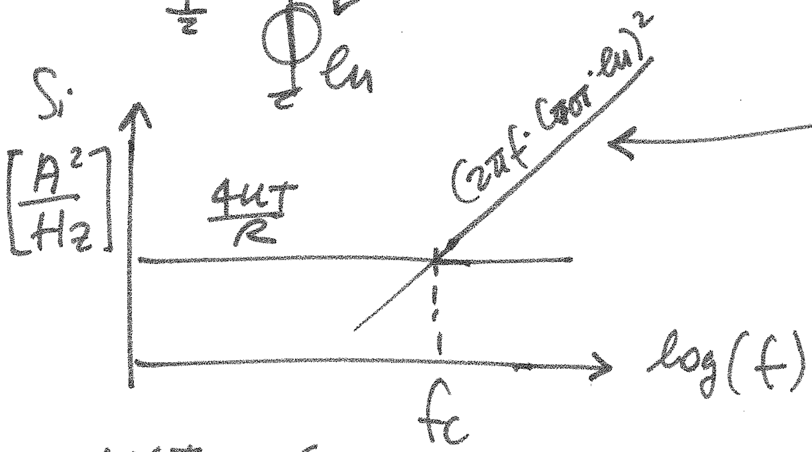
$$BW = \frac{1}{2\pi RC} \longrightarrow C = \frac{1}{2\pi R BW} = 15.7 \qquad = \cancel{101}\ \underset{25}{}\ kHz$$

$$= 16\ pF \qquad 4 \qquad 64$$

e)



$$S_i\Big|_{TOT} = \frac{4KT}{R} + (2\pi f \cdot C_{TOT} \cdot e_n)^2$$



$$C_{TOT} = C_{DL} + C =$$
$$= 210 + \cancel{16}\ \underset{64}{} = \cancel{226}\ \overset{74}{}\ pF$$

$$\frac{4KT}{R} = (2\pi f \cdot C_{TOT} \cdot e_n)^2 \Longrightarrow f_c = \frac{\sqrt{\frac{4KT}{R}}}{2\pi\, C_{TOT}\, e_n} = \frac{0.\cancel{4}pA/\sqrt{Hz}}{2\pi\, C_{TOT} \cdot e_n}$$

$$= \cancel{47}\ \overset{77}{}\ \cancel{kHz}$$

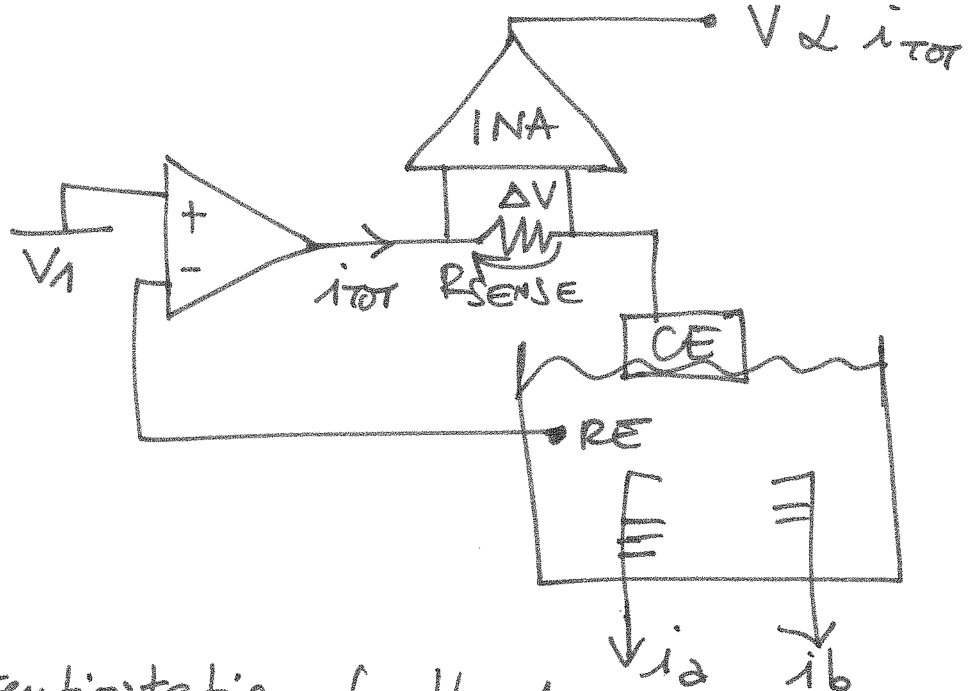
f) $i_{REDOX} \propto Area$

$i_{REDOX} \propto \sqrt{\text{SCAN RATE}}$

AREA $\downarrow$ 2 $\Longrightarrow$ SCAN RATE INCREASE by 4

g) W ~~is~~ Are the min. feature 5 μm ⟶ accuracy better than 5 μm
(and G) if WEa and WEb DIFFERENT METALS, otherwise SINGLE MASK

h)



The potentiostatic feedback loop compensates for the Ohmic drop ΔV. (amplified by the INA)

(COMMON MODE Check)